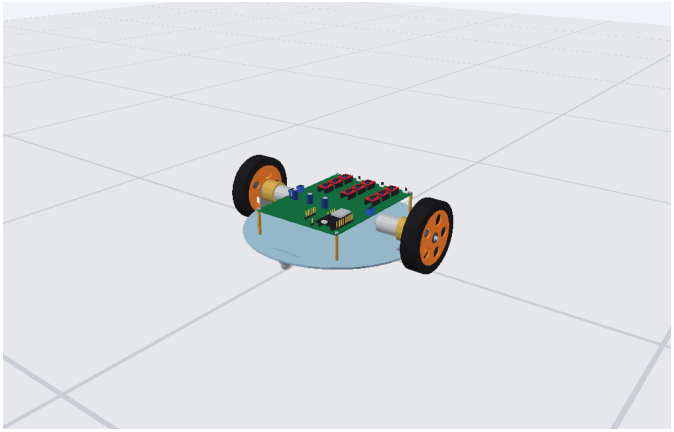
RocketBoy – specification sheet



Legend: [given] read from your files (STEP / PDFs / KiCad), [derived] calculated from them, [assumed] my choice – confirm or replace, [sim] simulated, not measured.

Base robot

Parts used

#	Part	Qty	Notes
1	Main PCB with header pins (RocketBoy.kicad_pcb)	1	155 × 208 × 1.6 mm, 2-layer, 2 oz copper recommended
2	ESP32 DevKit (38-pin)	1	plugs into socket U10
3	Johnson gear-motor SG-3851218000-60K, 12 V	2	
4	Custom 3D-printed / CNC chassis	1	Ø300 × 3 mm disc
5	L-clamp for Johnson motor	2	1.5 mm steel
6	TB6612FNG driver module	9	3 enable groups of 3
7	Wheel Ø110 × 25 mm, 6 mm bore [assumed]	2	
8	Ball caster, 25.3 mm tall [assumed]	2	front and rear
9	3S Li-ion pack (Molicel 18650, 2.8 Ah class) [assumed]	1	under the chassis
10	12 V -> 5 V buck module, >= 2 A [assumed]	1	wired to J1 / J2
11	SPDT slide switch	3	SW1–SW3, enable the 3 driver groups
12	470 µF / 25 V electrolytic capacitor	3	C1–C3 on the motor supply
13	5.08 mm screw terminal, 2-pole	3	battery, left motor, right motor
14	M3 × 40 mm standoff + M3 screws	4	PCB on the chassis holes (±75, ±100 mm)
15	Fuse 10 A + main switch [recommended]	1	in the battery lead (both motors jammed = 13 A)
16	Header pins J3 (1×10), J4 (1×16), J5 (1×8), 2.54 mm	3	for a top PCB / add-ons

Sizes

Item	Size
Overall (wheel to wheel × length × height)	~ 358 × 300 × 86 mm [derived]
Chassis	Ø300 mm × 3 mm [given]
Wheel	Ø110 × 25 mm [assumed]; track (wheel centre to centre) 331 mm
Chassis underside above floor	25.3 mm (top face 28.3 mm) [derived]
Motor	gearbox Ø37 × 25, can Ø29 × 37.8, shaft Ø6 × 30 mm [given]
L-clamp	40 mm wide, flange 22 mm, motor axis 26.7 mm above the flange [given]
PCB	155 × 208 × 1.6 mm, 4 × Ø3.2 mm holes on a 150 × 200 mm pattern [given]
Mass	~ 2.2 kg [assumed part list]

Specification

Item	Value
Supply	3S Li-ion, 9 – 12.6 V (nominal 11.1 V)
Logic	ESP32 on 5 V from the buck module; 3.3 V for the driver modules' logic
Motor	12 V DC, 300 rpm no-load, 220 rpm at 6 kgf·cm, no-load current <= 0.6 A [given]
Motor stall (not in the PDF)	7.5 A [given by you] at 12 V; ~ 25 kgf·cm [derived]; in the robot 6.1 – 6.5 A (pack / driver / wiring resistance)
Driver	TB6612FNG ×9 in parallel: 1.2 A continuous / 3.2 A pulse per channel, R_on ~ 0.5 ohm
Channels	A = left motor, B = right motor, same signals on all 9 modules
Driver capacity per motor	1 group 3.6 A · 2 groups 7.2 A · 3 groups 10.8 A continuous
Enable	STBY1–3 via slide switches SW1–SW3 (ON = 3.3 V, OFF = GND)
Control pins	PWMA GPIO13, AIN1 GPIO12, AIN2 GPIO14, PWMB GPIO27, BIN1 GPIO25, BIN2 GPIO26
Top speed	1.73 m/s [sim]
Acceleration	0 -> 1 m/s in 0.18 s [sim]
Spin on the spot	~ 272 °/s at duty ±0.5 [sim]
Slope	tyres slip at ~ 34° (mu 0.8 assumed) [sim]
Current	cruise ~ 0.5 A per motor; start-up spike ~ 7 A for ~50 ms (use soft start); jammed wheel ~ 6.1 – 6.5 A per motor, ~ 13 A from the battery for both [sim]
Runtime	~ 2 h at steady cruise on 2.8 Ah (~ 1.4 A total) [estimate]
Expansion	J3: 5V, 3V3, GND, BIN1, BIN2, PWMB, AIN2, AIN1, GND, PWMA · J4/J5: all usable free ESP32 pins

Line follower add-on

Parts used

#	Part	Qty	Notes
1	Base robot (above)	1	unchanged
2	8-channel IR reflectance array (Indian Hobby Center "Line Follower Sensor 8 Array")	1	pinout not verified (shop page blocked for me)
3	Sensor mounting bracket (3D-printed)	1	array 3–8 mm above the floor, at the front underside
4	Jumper wires / ribbon to J5 (and J4 if needed)	1 set	
5	Optional: top add-on PCB on J3 / J4 / J5	0–1	

Sizes

Item	Size
Sensor array	width ~ 70 mm for 8 sensors at 9.525 mm pitch [assumed, QTR-8 style] – check yours
Mounting	under the chassis front; chassis underside is 25.3 mm above the floor, so the bracket drops ~ 17–22 mm
Line	19 mm black tape on a light floor [assumed]

Specification

Item	Value
Sensor supply	3.3 V (J5 pin 1) so outputs never exceed the ESP32's 3.3 V limit
Analog inputs	J5: GPIO36, 39, 34, 35, 32, 33 (ADC1, work with Wi-Fi on)
Extra digital inputs	J4: e.g. GPIO18, GPIO19 for sensors 7 and 8
Control	PID on the weighted line position -> left/right duty (firmware not included)
Driver groups needed	1 group is enough; keep the duty slew limit (3 per second)
Performance	not simulated in the current simulator [open]